



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Year: 2026 Trimester: Spring

CSE 4165/CSE 465 Web Programming

Total Marks: 30

Duration: 2 Hours

SET - A

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1.	Develop a JavaScript application where users can track their weekly study hours by subject and get feedback. At first, create two input fields: Subject Name and Study Hours, an “Add Study” button, and a label area to display total study hours, average study hours, subjects studied, and feedback. The user will enter the subject name and the number of hours studied in each session. When the button is clicked, the application will do the following: a. Set a weekly study goal of 20 hours. b. Add the entered study hours to a running total. c. Store each entered subject name. d. Count how many study sessions the user has entered. e. Calculate the average study hour so far: Average = Total Study Hours / Number of Sessions f. Display: - o total study hours - o average study hours - o subjects studied in all sessions - o feedback message g. Display feedback based on total study hours: - o 0–5 → “Start studying regularly!” - o 6–12 → “Good progress!” - o 13–19 → “Almost ready!” - o 20+ → “Weekly goal reached!” h. If the user enters more than 7 study sessions without reaching 20 hours, display “Increase your study time!” <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>English-2, English-3, Bangla-4, Physics-5</td><td>Total: 14 hours, Average: 3.5 hours, Subjects: [English, English, Bangla, Physics], Almost ready!</td></tr><tr><td>Math-5, Physics-6, Chemistry-7, English-3</td><td>Total: 21 hours, Average: 5.25 hours, Subjects: [Math, Physics, Chemistry, English], Weekly goal reached!</td></tr><tr><td>8 sessions, total 15 hours</td><td>Total: 15 hours, Average: 1.875 hours, Increase your study time!</td></tr></table>	Sample Input	Sample Output	English-2, English-3, Bangla-4, Physics-5	Total: 14 hours, Average: 3.5 hours, Subjects: [English, English, Bangla, Physics], Almost ready!	Math-5, Physics-6, Chemistry-7, English-3	Total: 21 hours, Average: 5.25 hours, Subjects: [Math, Physics, Chemistry, English], Weekly goal reached!	8 sessions, total 15 hours	Total: 15 hours, Average: 1.875 hours, Increase your study time!	[10]
Sample Input	Sample Output									
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2.	Write a PHP program for a student academic tracking system. The program should take three inputs from the user via an HTML form: ● credits earned per course ● number of courses completed ● target graduation credits. First, calculate the total credits earned by multiplying the credits per course by the number of courses completed. Classify the academic progress using the following conditions: ● Excellent: total credits 120 or more ● Good: total credits 90 or more but less than 120 ● Average: total credits 60 or more but less than 90 ● Poor: total credits less than 60 The program should then compare the total credits earned with the target graduation credits. Finally, display the total credits, the progress category, the target value, and clearly state whether the progress is above, below, or exactly at the target, along with the difference.	[10]								

Sample:					
Credits/ Course	Courses Completed	Targ et	Total Credits	Progres s	Result
4	30	120	120	Excellen t	Target met exactly (0 difference)
3	35	100	105	Good	Above target by 5
3	25	90	75	Average	Below target by 15
4	10	60	40	Poor	Below target by 20

3.

Create a database named tourism and a table named tourist_spot with the following structure and values:

SpotID	SpotName	Region	Category	Rating	EntryFee	VisitorsPerYe ar
1	Cox's Bazar Beach	CTG	Beach	4.9	0	150000
2	Sundarbans Forest	KHL	Nature	4.3	150	80000
3	Ratargul Forest	SYL	Nature	4.8	100	60000
4	Kuakata Beach	PTU	Beach	4.5	0	90000
5	Shat Gambuj Mosque	BGH	Heritage	4.7	30	70000
6	Paharpur	NWG	Heritage	4.1	30	40000

Now, do the following (write full PHP–MySQL code):

1.

Display the SpotName, Region and Rating of all spots whose rating is above 4.5 or whose SpotName contains the word 'Beach'. Sort the result by Rating in descending order.

2.

Update: for all spots whose rating is between 4.0 and 4.5 (inclusive) and whose EntryFee is greater than 0, increase the Rating by 0.2 and raise the EntryFee by 10%.

3.

For each category, show the category name, the number of spots in it, the average rating, and the total visitors per year but only include categories whose average rating is above 4.4. Sort the result by average rating in descending order.

4.

For each region, show the region name and its total estimated yearly revenue, calculated as the sum of (EntryFee × VisitorsPerYear) over its spots. Display only those regions whose total estimated revenue exceeds 1,000,000, ordered from highest to lowest.

[10]



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SET - B

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

1.	Create a JavaScript program where a user enters a patient's vitals and your code shows a live health report. Strategies are as follows: Index.html: Create two text fields (input type text) to take the user's Heart Rate (bpm) and Oxygen Saturation / SpO2 (%), and a "Save Reading" button. Below the button there is a label field to show the report and feedback. Script.js: 1. Take the user's Heart Rate and SpO2 from the input fields in the HTML file. 2. Keep track of every reading entered so far (for example, in arrays). 3. Inside a function: a) Calculate and display the Average Heart Rate (of all readings entered so far, including the current one). b) Calculate and display the Average Oxygen Saturation (of all readings so far, including the current one). c) Calculate and display the Risk Score from the current reading using the formula: $\text{RiskScore} = (100 - \text{SpO2}) \times 2 + \text{HeartRate} - 80 \times 0.5$<i>(HeartRate - 80 is the absolute / positive difference, e.g. 78 - 80 = 2.)</i> d) Display the feedback in the label area: "SAFE" if the Risk Score is 10 or less, "WARNING" if it is greater than 10 and 20 or less, and "DANGER" if it is greater than 20. Sample Input / Output <table><thead><tr><th>Sample Input</th><th colspan="2">Sample Output</th></tr><tr><th>Heart Rate, SpO2 (entered in order)</th><th>Report</th><th>Feedback</th></tr></thead><tbody><tr><td>80, 97</td><td>Average Heart Rate: 80 Average Oxygen Saturation: 97 Risk Score: 6.0</td><td>SAFE</td></tr><tr><td>78, 95</td><td>Average Heart Rate: 79 Average Oxygen Saturation: 96 Risk Score: 11</td><td>WARNING</td></tr><tr><td>130, 80</td><td>Average Heart Rate: 96 Average Oxygen Saturation: 90.67 Risk Score: 65</td><td>DANGER</td></tr></tbody></table>	Sample Input	Sample Output		Heart Rate, SpO2 (entered in order)	Report	Feedback	80, 97	Average Heart Rate: 80 Average Oxygen Saturation: 97 Risk Score: 6.0	SAFE	78, 95	Average Heart Rate: 79 Average Oxygen Saturation: 96 Risk Score: 11	WARNING	130, 80	Average Heart Rate: 96 Average Oxygen Saturation: 90.67 Risk Score: 65	DANGER	[10]
Sample Input	Sample Output																
Heart Rate, SpO2 (entered in order)	Report	Feedback															
80, 97	Average Heart Rate: 80 Average Oxygen Saturation: 97 Risk Score: 6.0	SAFE															
78, 95	Average Heart Rate: 79 Average Oxygen Saturation: 96 Risk Score: 11	WARNING															
130, 80	Average Heart Rate: 96 Average Oxygen Saturation: 90.67 Risk Score: 65	DANGER															
2.	The UIU Mars Rover Team is preparing for a rover field-test mission and needs to calculate how many battery modules must be carried for the rover. Write a PHP program that takes three inputs from an HTML form: the mission distance in meters, the energy consumption per meter, and the capacity of one battery module. The program should first calculate the total energy required by multiplying the mission distance with the energy consumption per meter. Since the team cannot carry partial battery modules, the program must calculate the minimum number of complete battery modules needed. Then, calculate the unused energy capacity after the required number of battery modules is carried. Classify the battery usage efficiency using the following conditions: • Efficient: unused energy is less than or equal to 10% of total carried energy • Acceptable: unused energy is more than 10% but less than or equal to 25% of total carried energy	[10]															

- **Wasteful:** unused energy is more than 25% of total carried energy

Finally, display the **total energy required**, **number of battery modules needed**, **unused energy capacity**, and the **battery usage status**.

Sample Input taken using HTML form			Sample Output printed using PHP code			
Mission Distance	Energy per Meter	Battery Capacity	Total Energy Required	Battery Modules	Unused Energy	Status
1200	8	5000	9600	2	400	Efficient
1500	9	4000	13500	4	2500	Acceptable
1000	6	5000	6000	2	4000	Wasteful

3. Create a database named 'uiu_bookshop' and a table named 'book_info' with the following structure and values:

[10]

BookID	BookTitle	Author	Category	Price	Stock
1	Web Basics	John Smith	Programming	450	10
2	Database Guide	Alice Roy	Programming	550	8
3	English Grammar	Mary Khan	Language	300	15
4	Business Math	David Lee	Business	400	12
5	Python Mastery	Sara Ahmed	Programming	600	5
6	French Made Easy	Paul Costa	Language	350	7

Now, do the following (**write full PHP–MySQL code**):

1. Display all books from the Programming category, sorted by Price in descending order.
2. Display the BookTitle, Author, and Price of all books where the Price is greater than 400 BDT and the Category is 'Programming'.
3. Display the number of books available in each Category.
4. Display the total net worth of the entire bookshop. (Hint: To find the total net worth of the entire bookshop, multiply the Price and Stock of each book, then add all the values together.)
5. Apply a 10% price reduction to all books in the 'Language' category, then display the BookTitle and updated Price of those books.